



S3 API compatibility

R2 implements the S3 API to allow users and their applications to migrate with ease. When comparing to AWS S3, Cloudflare has removed some API operations' features and added others. The S3 API operations are listed below with their current implementation status. Feature implementation is currently in progress. Refer back to this page for updates. The API is available via the

`https://<ACCOUNT_ID>.r2.cloudflarestorage.com` endpoint. Find your [account ID in the Cloudflare dashboard](#).

How to read this page

This page has two sections: bucket-level operations and object-level operations.

Each section will have two tables: a table of implemented APIs and a table of unimplemented APIs.

Refer the feature column of each table to review which features of an API have been implemented and which have not.



Feature Implemented



Feature Implemented (Experimental)



Feature Not Implemented

Bucket region

When using the S3 API, the region for an R2 bucket is auto. For compatibility with tools that do not allow you to specify a region, an empty value and `us-east-1` will alias to the auto region.

This also applies to the `LocationConstraint` for the `CreateBucket` API.

Checksum Types

Checksums have an algorithm and a [type](#). Refer to the table below.

Checksum Algorithm	FULL_OBJECT	COMPOSITE
CRC-64/NVME (CRC64NVME)		
CRC-32 (CRC32)		
CRC-32C (CRC32C)		
SHA-1 (SHA1)		
SHA-256 (SHA256)		

Bucket-level operations

The following tables are related to bucket-level operations.

Implemented bucket-level operations

Below is a list of implemented bucket-level operations. Refer to the Feature column to review which features have been implemented (✅) and have not been implemented (❌).

API Name	Feature
✅ ListBuckets ↗	
✅ HeadBucket ↗	❌ Bucket Owner: ❌ x-amz-expected-bucket-owner
✅ CreateBucket ↗	❌ ACL: ❌ x-amz-acl

API Name

Feature

 x-

amz-grant-
full-control

 x-

amz-grant-
read

 x-

amz-grant-
read-acp

 x-

amz-grant-
write

 x-

amz-grant-
write-acp

 Object

Locking:

 x-

amz-
bucket-

API Name	Feature
	object-lock-enabled  Bucket Owner:  x-amz-expected-bucket-owner
 DeleteBucket ↗	 Bucket Owner:  x-amz-expected-bucket-owner
 DeleteBucketCors ↗	 Bucket Owner:

API Name	Feature
	✗ x- amz- expected- bucket- owner
✓ GetBucketCors ↗	✗ Bucket Owner: ✗ x- amz- expected- bucket- owner

API Name

Feature

[GetBucketLifecycleConfiguration ↗](#) Bucket

Owner:

 x-

amz-

expected-

bucket-

owner

[GetBucketLocation ↗](#) Bucket

Owner:

 x-

amz-

expected-

bucket-

owner

[GetBucketEncryption ↗](#) Bucket

Owner:

 x-amz-

expected-

API Name	Feature
	bucket-owner
 PutBucketCors ↗	 Checksums:  x-amz-sdk-checksum-algorithm  x-amz-checksum-algorithm  Bucket Owner:  x-amz-expected-bucket-owner

API Name

Feature



[PutBucketLifecycleConfiguration ↗](#)



Checksums:



x-

amz-sdk-

checksum-

algorithm



x-

amz-

checksum-

algorithm



Bucket

Owner:



x-

amz-

expected-

bucket-

owner

Unimplemented bucket-level operations

> Unimplemented bucket-level operations

Object-level operations

The following tables are related to object-level operations.

Implemented object-level operations

Below is a list of implemented object-level operations. Refer to the Feature column to review which features have been implemented (✅) and have not been implemented (❌).

API Name	Feature
✅ HeadObject ↗	✅ Conditional Operations:

API Name

Feature

 If-Match

 If-

Modified-Since

 If-None-

Match

 If-

Unmodified-

Since

 Range:

 Range (has
no effect in
HeadObject)



partNumber

 SSE-C:

 x-amz-

server-side-

encryption-

customer-

algorithm

API Name

Feature



x-amz-server-side-encryption-customer-key



x-amz-server-side-encryption-customer-key-MD5



Request

Payer:



x-amz-request-payer



Bucket

Owner:



x-amz-expected-bucket-owner

API Name

Feature

 [ListObjects ↗](#)

Query

Parameters:

 delimiter

 encoding-

type

 marker

 max-keys

 prefix

 Request

Payer:

 x-amz-

request-payer

 Bucket

Owner:

 x-amz-

expected-bucket-owner

 [ListObjectsV2 ↗](#)

Query

Parameters:

API Name

Feature

 list-type



continuation-
token

 delimiter

 encoding-
type

 fetch-

owner

 max-keys

 prefix

 start-after

 Request

Payer:

 x-amz-

request-payer

 Bucket

Owner:

 x-amz-

API Name

Feature

expected-bucket-owner

 [GetObject](#) ↗

 Conditional

Operations:

 If-Match

 If-

Modified-Since

 If-None-

Match

 If-

Unmodified-

Since

 Range:

 Range



PartNumber

 SSE-C:

 x-amz-

server-side-

API Name

Feature

	encryption-
	customer-
	algorithm
	 x-amz-
	server-side-
	encryption-
	customer-key
	 x-amz-
	server-side-
	encryption-
	customer-key-
	MD5
	 Request
	Payer:
	 x-amz-
	request-payer
	 Bucket
	Owner:
	 x-amz-

API Name	Feature
	expected-bucket-owner
 PutObject 	 System Metadata:  Content-Type  Cache-Control  Content-Disposition  Content-Encoding  Content-Language  Expires  Content-MD5  Storage Class:  x-amz-

API Name

Feature

storage-class



STANDARD



STANDARD_IA

✗ Object

Lifecycle

✗ Website:

✗ x-amz-

website-redirect-
location

✗ SSE:

✗ x-amz-

server-side-
encryption-aws-
kms-key-id

✗ x-amz-

server-side-
encryption

✗ x-amz-

API Name

Feature

server-side-
encryption-
context

 x-amz-

server-side-
encryption-
bucket-key-
enabled

 SSE-C:

 x-amz-

server-side-
encryption-
customer-
algorithm

 x-amz-

server-side-
encryption-
customer-key

 x-amz-

server-side-

API Name

Feature

encryption-
customer-key-
MD5

✗ Request
Payer:

✗ x-amz-
request-payer

✗ Tagging:

✗ x-amz-
tagging

✗ Object
Locking:

✗ x-amz-
object-lock-
mode

✗ x-amz-
object-lock-
retain-until-date

✗ x-amz-
object-lock-

API Name

Feature

legal-hold

✗ ACL:

✗ x-amz-acl

✗ x-amz-

grant-full-control

✗ x-amz-

grant-read

✗ x-amz-

grant-read-acp

✗ x-amz-

grant-write-acp

✗ Bucket

Owner:

✗ x-amz-

expected-bucket-owner

✓ [DeleteObject](#) ↗

✗ Multi-factor authentication:

✗ x-amz-mfa

API Name

Feature

✗ Object

Locking:

✗ x-amz-

bypass-

governance-

retention

✗ Request

Payer:

✗ x-amz-

request-payer

✗ Bucket

Owner:

✗ x-amz-

expected-bucket-

owner

✓ [DeleteObjects](#) ↗

✗ Multi-factor
authentication:

✗ x-amz-mfa

✗ Object

API Name

Feature

Locking:

 x-amz-

bypass-

governance-

retention

 Request

Payer:

 x-amz-

request-payer

 Bucket

Owner:

 x-amz-

expected-bucket-
owner

 [ListMultipartUploads](#) ↗

 Query

Parameters:

 delimiter

 encoding-

type

API Name

Feature

	✓ key-marker ✓ max-uploads ✓ prefix ✓ upload-id-marker
--	--

✓

[CreateMultipartUpload ↗](#)

✓

System Metadata:

✓

Content-Type

✓

Cache-Control

✓

Content-Disposition

✓

Content-Encoding

✓

Content-Language

API Name

Feature

 Expires

 Content-

MD5

 Storage Class:

 x-amz-

storage-class



STANDARD



STANDARD_IA

 Website:

 x-amz-

website-redirect-
location

 SSE:

 x-amz-

server-side-
encryption-aws-
kms-key-id

 x-amz-

API Name

Feature

server-side-
encryption

 x-amz-

server-side-
encryption-
context

 x-amz-

server-side-
encryption-
bucket-key-
enabled

 SSE-C:

 x-amz-

server-side-
encryption-
customer-
algorithm

 x-amz-

server-side-
encryption-

API Name

Feature

customer-key



x-amz-

server-side-

encryption-

customer-key-

MD5



Request

Payer:



x-amz-

request-payer



Tagging:



x-amz-

tagging



Object

Locking:



x-amz-

object-lock-

mode



x-amz-

object-lock-

API Name

Feature

retain-until-date

✗ x-amz-

object-lock-

legal-hold

✗ ACL:

✗ x-amz-acl

✗ x-amz-

grant-full-control

✗ x-amz-

grant-read

✗ x-amz-

grant-read-acp

✗ x-amz-

grant-write-acp

✗ Bucket

Owner:

✗ x-amz-

expected-bucket-owner

API Name

Feature



[CompleteMultipartUpload ↗](#)

✗ Bucket

Owner:

✗ x-amz-

expected-bucket-owner

✗ Request

Payer:

✗ x-amz-

request-payer



[AbortMultipartUpload ↗](#)

✗ Request

Payer:

✗ x-amz-

request-payer



[CopyObject ↗](#)

✓ Operation

Metadata:

✓ x-amz-

metadata-directive

API Name

Feature

	 System Metadata:  Content-Type  Cache-Control  Content-Disposition  Content-Encoding  Content-Language  Expires  Conditional Operations:  x-amz-copy-source  x-amz-copy-source-if-match
--	---

API Name

Feature



x-amz-

copy-source-if-
modified-since



x-amz-

copy-source-if-
none-match



x-amz-

copy-source-if-
unmodified-since



Storage Class:



x-amz-

storage-class



STANDARD



STANDARD_IA



ACL:



x-amz-acl



x-amz-

grant-full-control

API Name

Feature

✗ x-amz-grant-read

✗ x-amz-grant-read-acp

✗ x-amz-grant-write-acp

✗ Website:

✗ x-amz-website-redirect-location

✗ SSE:

✗ x-amz-server-side-encryption

✗ x-amz-server-side-encryption-aws-kms-key-id

✗ x-amz-server-side-

API Name

Feature

	encryption-context
	 x-amz-server-side-encryption-bucket-key-enabled
	 SSE-C:
	 x-amz-server-side-encryption-customer-algorithm
	 x-amz-server-side-encryption-customer-key
	 x-amz-server-side-encryption-

API Name

Feature

customer-key-MD5

 x-amz-

copy-source-server-side-encryption-customer-algorithm

 x-amz-

copy-source-server-side-encryption-customer-key

 x-amz-

copy-source-server-side-encryption-customer-key-MD5

 Request

API Name

Feature

Payer:

✗ x-amz-request-payer

✗ Tagging:

✗ x-amz-tagging

✗ x-amz-tagging-directive

✗ Object

Locking:

✗ x-amz-object-lock-mode

✗ x-amz-object-lock-retain-until-date

✗ x-amz-object-lock-legal-hold

✗ Bucket

API Name

Feature

Owner:

✗ x-amz-

expected-bucket-owner

✗ x-amz-

source-expected-bucket-owner

✗ Checksums:

✗ x-amz-

checksum-algorithm

✓ [UploadPart ↗](#)

✓ System

Metadata:

✓ Content-

MD5

✗ SSE:

✗ x-amz-

server-side-encryption

API Name

Feature



SSE-C:



x-amz-server-side-encryption-customer-algorithm



x-amz-server-side-encryption-customer-key



x-amz-server-side-encryption-customer-key-MD5



Request

Payer:



x-amz-request-payer



Bucket

API Name

Feature

Owner:

✗ x-amz-

expected-bucket-owner

✓ [UploadPartCopy ↗](#)

✗ Conditional Operations:

✗ x-amz-

copy-source

✗ x-amz-

copy-source-if-match

✗ x-amz-

copy-source-if-modified-since

✗ x-amz-

copy-source-if-none-match

✗ x-amz-

copy-source-if-

API Name

Feature

unmodified-since

✓ Range:

✓ x-amz-

copy-source-
range

✓ SSE-C:

✓ x-amz-

server-side-
encryption-
customer-
algorithm

✓ x-amz-

server-side-
encryption-
customer-key

✓ x-amz-

server-side-
encryption-
customer-key-
MD5

API Name

Feature

 x-amz-copy-source-server-side-encryption-customer-algorithm

 x-amz-copy-source-server-side-encryption-customer-key

 x-amz-copy-source-server-side-encryption-customer-key-MD5

 Request Payer:

 x-amz-

API Name

Feature

request-payer

✗ Bucket

Owner:

✗ x-amz-

expected-bucket-owner

✗ x-amz-

source-expected-bucket-owner



[ListParts](#) ↗

Query

Parameters:

✓ max-parts

✓ part-

number-marker

✗ Request

Payer:

✗ x-amz-

request-payer

✗ Bucket

API Name

Feature

Owner:

 x-amz-

expected-bucket-owner



Warning

Even though `ListObjects` is a supported operation, it is recommended that you use `ListObjectsV2` instead when developing applications. For more information, refer to [ListObjects ↗](#).

Unimplemented object-level operations

› Unimplemented object-level operations

Was this helpful?



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